

★ RS003 Week 10 — English Worksheet

Topic: Feedback When You Hit the Edge · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week is about **thinking and talking** about the code that makes the screen react when the ship hits a wall. The border flashes red and the scene shakes for a moment using **timer variables** that count down frame by frame and **random** to make a shake. You will read code, find bugs on paper, and explain the code you wrote in the lesson.

Words to know: **timer**, **flash**, **shake**, **offset**, **random**

1 · Predict 🧠

Read each piece of code. Write what you think it does. You are reading only — do not run anything.

```
flash_timer = 0
if ship_x > WIDTH - 20:
    ship_x = WIDTH - 20
    flash_timer = 10
```

When the ship hits the right wall, **flash_timer** is set to 10. What might 10 mean here?


```
if flash_timer > 0:
    flash_timer = flash_timer - 1
```

This runs every loop. How long does **flash_timer** stay above 0 after being set to 10?


```
shake_x = random.randint(-5, 5) if shake_timer > 0 else 0
pygame.draw.rect(screen, CYAN, (ship_x - 20 + shake_x, ship_y, 40, 50))
```

While the shake timer is running, `shake_x` is a small random number. What does adding it to every position do?


```
border_color = (255, 50, 50) if flash_timer > 0 else (60, 60, 90)
border_width = 6 if flash_timer > 0 else 2
```

What colour and width is the border while `flash_timer` is above 0? What about when it is 0?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Find the problem on paper.

Bug A — The border should flash for a moment then go back to grey, but once it flashes it stays red forever.

```
if ship_x > WIDTH - 20:  
    ship_x = WIDTH - 20  
    flash_timer = 10
```

Hint: the timer is set but never counts down.

Write the fixed code (show where the timer goes down):

Why was it wrong? Why does your fix work?

Bug B — The screen should only shake when you hit a wall, but it shakes all the time even in the middle.

```
shake_x = random.randint(-5, 5)
```

Hint: only shake while the shake timer is running.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The shake should move the whole ship together, but right now the nose jumps around while the body stays still, so they come apart.

```
pygame.draw.rect(screen, CYAN, (ship_x - 20, ship_y, 40, 50))
pygame.draw.polygon(screen, WHITE, [(ship_x + shake_x, ship_y - 20), (ship_x - 20 +
shake_x, ship_y), (ship_x + 20 + shake_x, ship_y)])
```

Hint: the body does not have the shake offset added.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working example. It is the full program from your lesson. Answer the questions by reading the code — you do not need to run it.

```

import pygame
import random
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

ship_x = CENTER_X
ship_y = HEIGHT - 100
SPEED = 6
flash_timer = 0
shake_timer = 0

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]: ship_x -= SPEED
    if keys[pygame.K_RIGHT]: ship_x += SPEED
    if keys[pygame.K_UP]: ship_y -= SPEED
    if keys[pygame.K_DOWN]: ship_y += SPEED

    if ship_x < 20 or ship_x > WIDTH - 20 or ship_y < 20 or ship_y > HEIGHT - 60:
        flash_timer = 10
        shake_timer = 10
    if ship_x < 20: ship_x = 20
    if ship_x > WIDTH - 20: ship_x = WIDTH - 20
    if ship_y < 20: ship_y = 20
    if ship_y > HEIGHT - 60: ship_y = HEIGHT - 60

    shake_x = random.randint(-5, 5) if shake_timer > 0 else 0
    shake_y = random.randint(-5, 5) if shake_timer > 0 else 0

    screen.fill((10, 12, 40))
    border_color = (255, 50, 50) if flash_timer > 0 else (60, 60, 90)
    border_width = 6 if flash_timer > 0 else 2
    pygame.draw.rect(screen, border_color, (0, 0, WIDTH, HEIGHT), border_width)
    pygame.draw.rect(screen, (80, 220, 255), (ship_x - 20 + shake_x, ship_y +
shake_y, 40, 50))
    pygame.draw.polygon(screen, (255, 255, 255), [(ship_x + shake_x, ship_y - 20 +

```

```

shake_y), (ship_x - 20 + shake_x, ship_y + shake_y), (ship_x + 20 + shake_x, ship_y
+ shake_y)])
    pygame.display.flip()

    if flash_timer > 0:
        flash_timer = flash_timer - 1
    if shake_timer > 0:
        shake_timer = shake_timer - 1

    clock.tick(60)

pygame.quit()

```

The line `if ship_x < 20 or ship_x > WIDTH - 20 ...` checks four things. What is it checking for?

When that check is true, which two timers get set, and what number do they get?

The four lines like `if ship_x < 20: ship_x = 20` keep the ship on screen. In your own words, what do they stop happening?

Why is `shake_x` added to both the rectangle and the polygon, instead of just one of them?

Near the bottom, both timers go down by 1 each loop. What would happen if you deleted those two `if` blocks?

4 · Explain Your Lesson Code 🎥

Explain the code **you** wrote in today's lesson. Record a short video on your phone. You can show the program running, then talk through your code like you are teaching someone. Try to use these words: **timer, flash, shake, offset, random.**

1. Fly into a wall and show the flash and shake.
2. Read your timer line out loud and say what 10 means.
3. Point at where the timer counts down each loop.
4. Show where the **offset** (shake) is added to the ship.

Write what you will say in your video. Plan it here before you record.

Submit ☒

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.